BANK CUSTOMER CHURN ANALYSIS

Predicting Customer Attrition Using Machine Learning and Data Analytics

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PRESENTATION OUTLINE

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03 Data Preparation

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BUSINESS PROBLEM

Background

Customer churn is a major challenge in banking because losing existing customers reduces revenue and increases acquisition costs

Problem Statement

The bank faces customer attrition and requires a data-driven solution to identify at-risk customers and understand the factors driving churn

Project Objectives

- Identify key drivers of customer churn.
- Analyze customer behavior and engagement patterns.
- Develop predictive models to identify customers at risk of churn between 2018 -2022
- Provide actionable recommendations to improve customer retention.

Dataset Size: 10,000

Observed Churn Rate: 20.38%

DATASET OVERVIEW

Metric	Value
Total Customers	10,000
Total Variables	21
Target Variable	Exited (Churn)
Churn Rate	20.38%

The dataset contains customer demographic, financial, engagement, and service-related information used to analyze and predict churn behavior.

KEY VARIABLES

Financial Variables

- Balance
- Number of Products

Behavioral Variables

- Activity Status
- Complaint Status
- Satisfaction Score

Demographic Variables

- Age
- Geography

DATA PREPARATION

Raw Dataset → Data Cleaning → Feature Engineering → Data Transformation → EDA & Machine Learning

Data Cleaning

- Checked for missing values
- Verified data types

Feature Engineering

- Created Age Group
- Created Tenure Group

Data Transformation

- Applied One-Hot Encoding to categorical variables

MACHINE LEARNING MODELS

Model Development

- Target Variable: Exited (Churn)
- Data split into Training and Testing sets.
- Class imbalance handled using appropriate techniques.
- Multiple classification algorithms were evaluated.
- Model performance assessed using Accuracy, Precision, Recall, and F1-Score.

Models Evaluated

- Logistic Regression
- Random Forest

Goal

- Identify customers likely to churn.
- Compare model performance.
- Select the best-performing model.

MODELING APPROACH

Objective

- Predict whether a customer will churn.
- Identify the key factors driving customer attrition.
- Compare model performance under different scenarios.

Models Developed

- Logistic Regression
- Random Forest

Modeling Scenarios

- Model 1: Included Complaint Status.
- Model 2: Excluded Complaint Status.

Reason for Two Models

- Complaint Status is a strong predictor of churn.
- A second model was developed without complaints to evaluate churn prediction using earlier customer behavior indicators.

MODEL PERFORMANCE COMPARISON

Final Model Comparison Table

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression (With Complain)	99.85%	99.75%	99.51%	99.63%
Logistic Regression (Without Complain)	81.30%	62.32%	21.08%	31.50%
Random Forest (With Complain)	99.85%	99.75%	99.51%	99.63%
Random Forest (Without Complain)	86.80%	79.51%	47.55%	59.51%

Key Findings

- Models including **Complaint Status** achieved the highest predictive performance (99.85% Accuracy, 99.63% F1 Score).

- Removing **Complaint Status** reduced model performance, highlighting its importance in predicting churn.

- **Random Forest (Without Complaint)** outperformed **Logistic Regression (Without Complaint)** across all evaluation metrics.

- Random Forest improved Recall from **21.08%** to **47.55%**, identifying substantially more churners.

Insight

Customer complaints are the strongest indicator of churn. However, the Random Forest model can effectively predict churn using demographic and behavioral factors, enabling earlier intervention before complaints occur.

BEST MODEL ANALYSIS

Best Performing Models

- Logistic Regression (With Complaint) achieved 99.85% accuracy.

- Random Forest (With Complaint) achieved 99.85% accuracy.

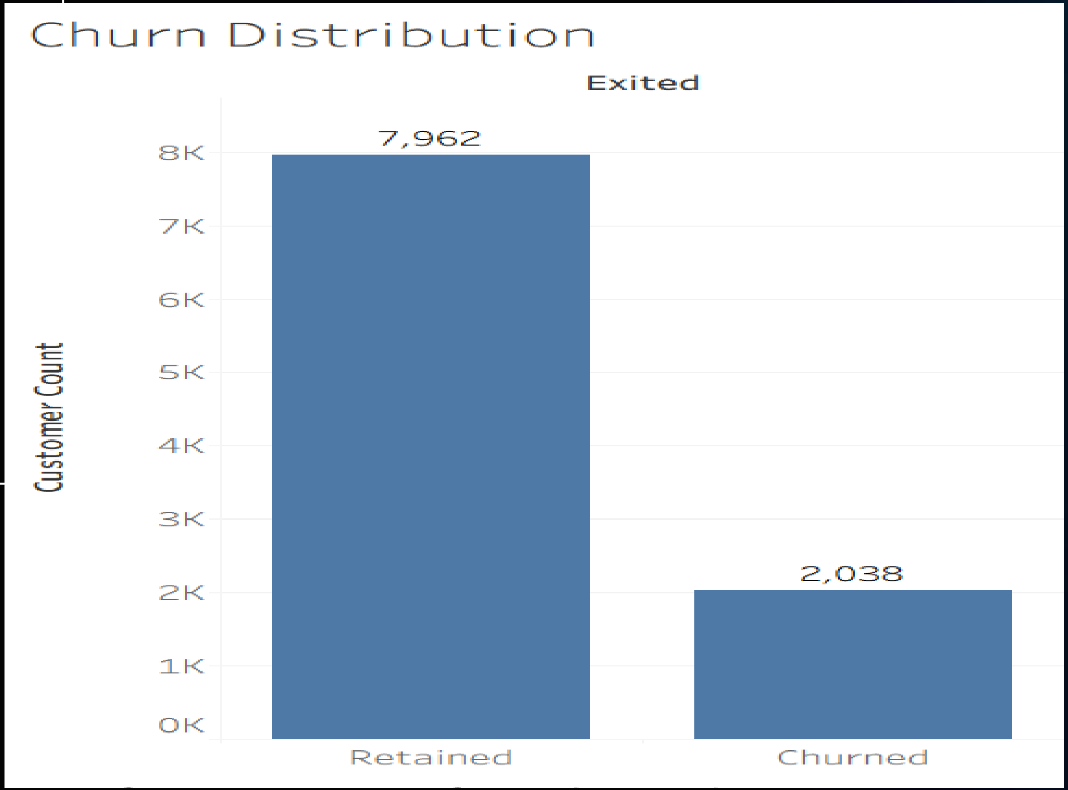
Best Practical Model

- Random Forest (Without Complaint): **86.80%Accuracy**
- Precision: **79.51%**
- Recall: **47.55%**
- F1 Score: **59.51%**

Why This Model?

- Predicts churn without relying on complaint information.
- Uses customer demographic and behavioral characteristics.
- Identifies at-risk customers before complaints occur.
- Supports proactive customer retention strategies.

CUSTOMER CHURN OVERVIEW

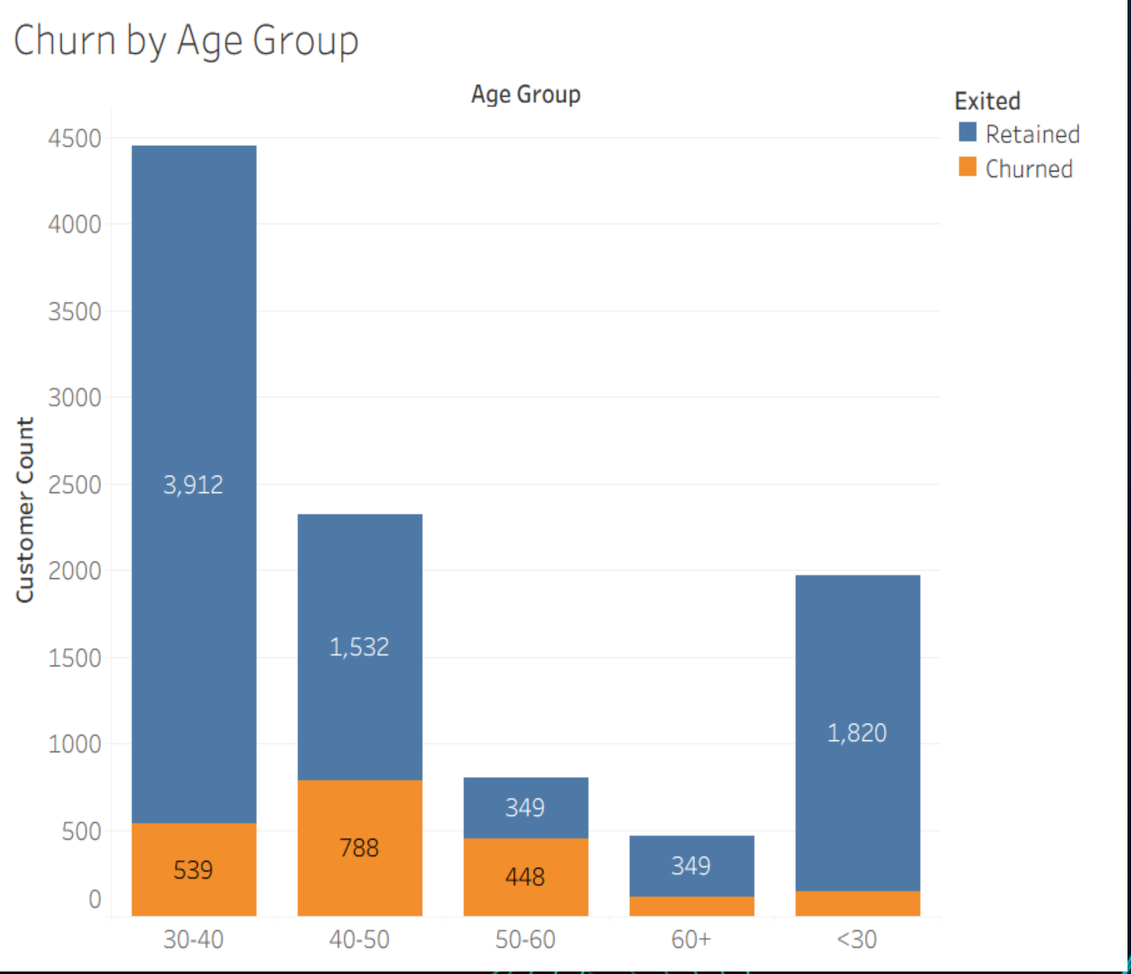


Key Findings

- Total Customers: **10,000**
- Churn Rate: **20.38%**
- Retained Customers: **7,962**
- Churned Customers: **2,038**
- Approximately 1 in 5 customers left the bank
- Retention Rate: **79.62%**

Insight: Although most customers were retained, the bank lost over 2,000 customers, highlighting the need to identify factors driving churn.

CHURN BY AGE GROUP



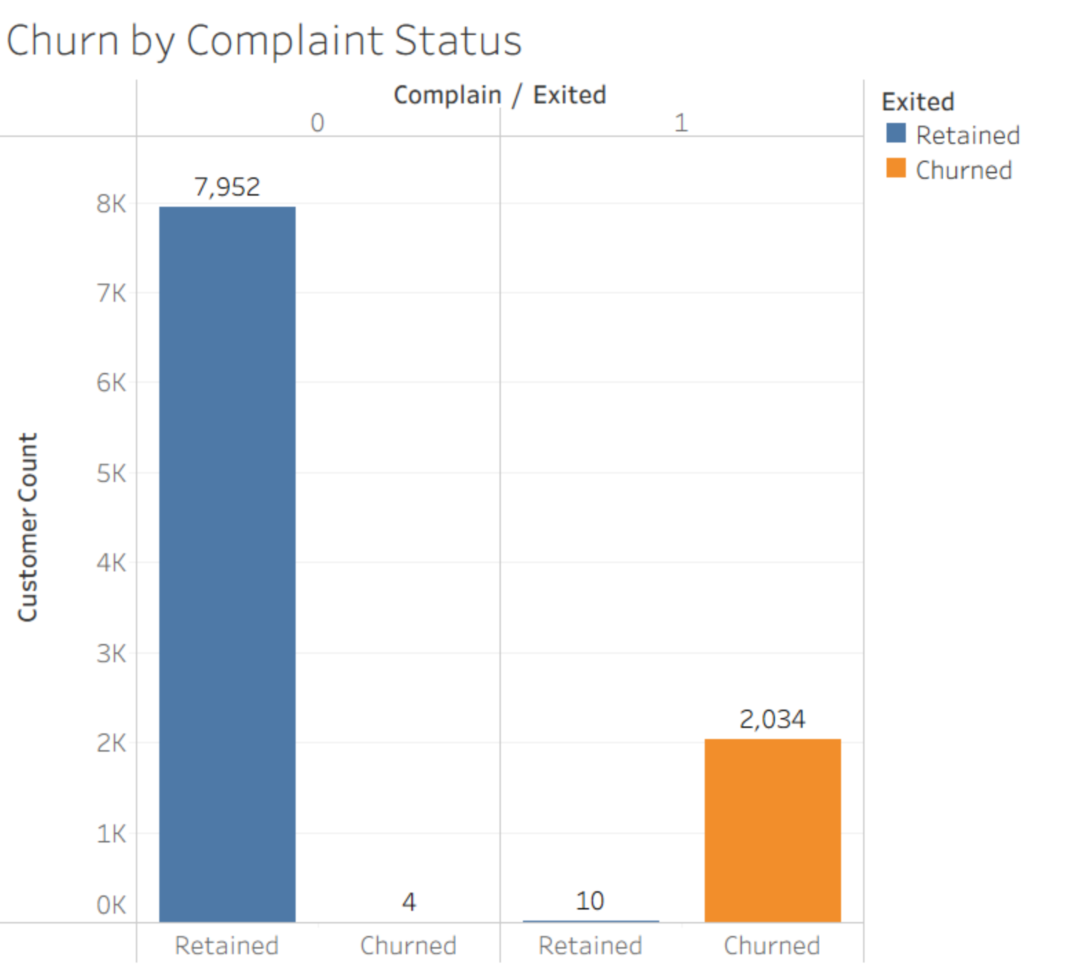
Key Findings

- Customers aged 50–60 exhibit a relatively high churn proportion
- Customers **under 30** show the strongest retention.
- Churn tendency generally increases with age.

Insight:

Older customers show higher churn rates, indicating that retention efforts should focus on customers aged 40 and above.

CHURN BY COMPLAINT STATUS



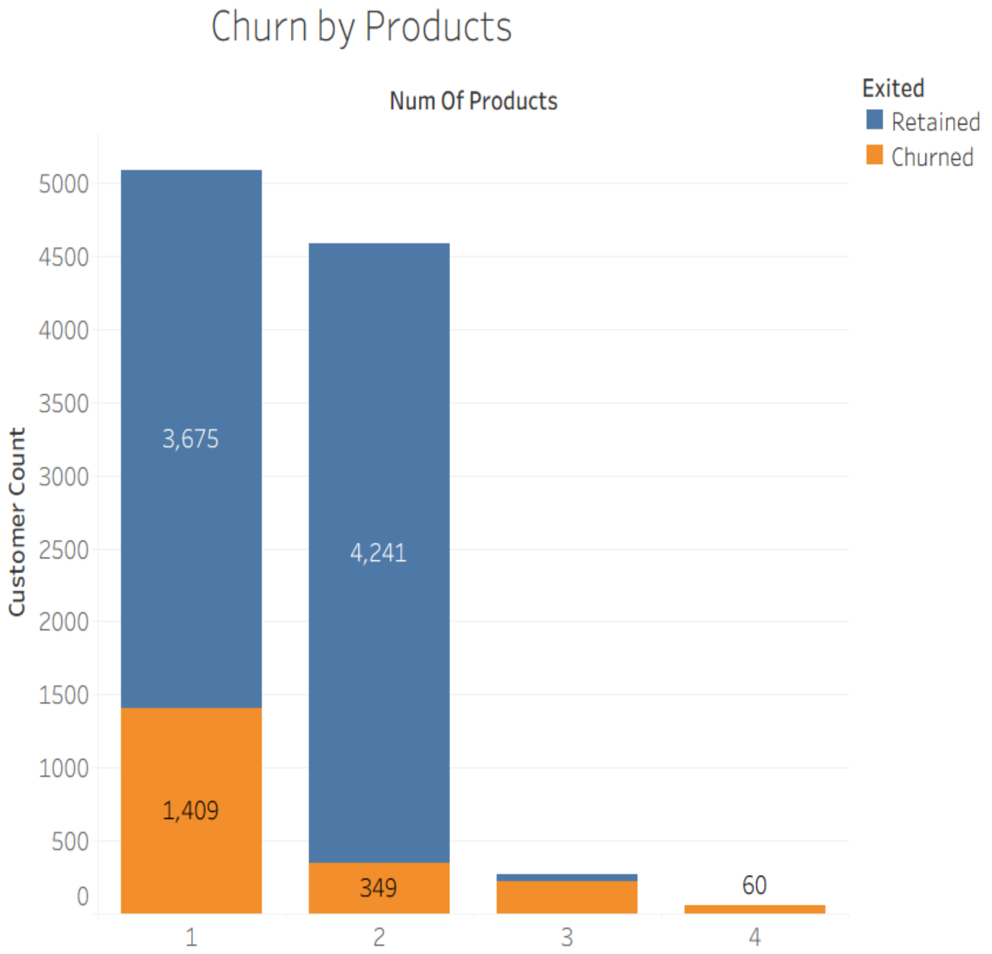
Key Findings

- Approximately 79.6% of customers were retained, while 20.4% churned.
- Nearly all churned customers are associated with complaint records.
- Customers without complaints demonstrate significantly higher retention levels.
- Complaint behavior shows a strong relationship with customer churn.

Insight

Customer dissatisfaction is a major driver of churn, making complaint management a critical area for retention strategies.

CHURN BY NUMBER OF PRODUCTS



Key Findings

- Customers with **1 product** recorded the highest churn count (**1,409 customers**).

- Customers with **2 products** exhibited the strongest retention (**4,241 retained customers**).

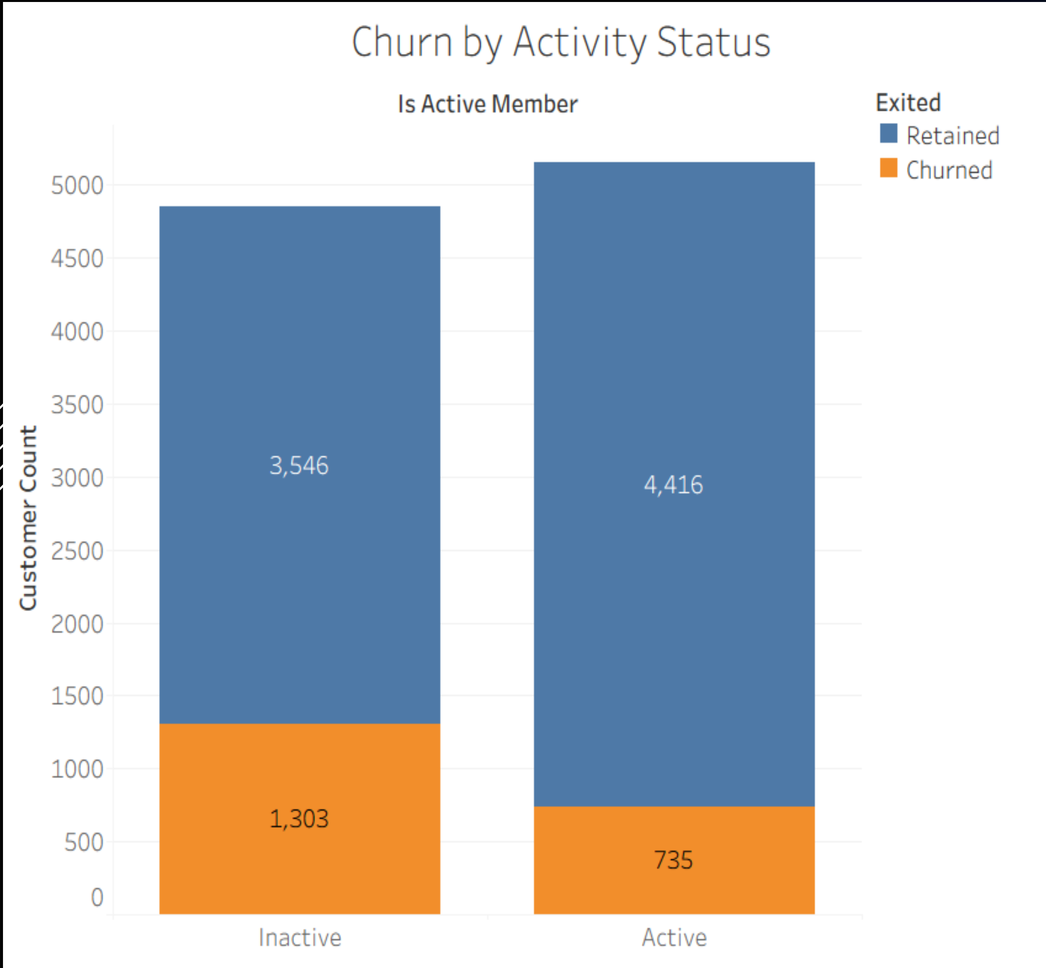
- Customers holding **3 or 4 products** represent a relatively small segment of the customer base.

- Churn generally decreases as the number of products owned increases.

Insight

Customers with multiple banking products tend to be more loyal and less likely to churn. Expanding product adoption through cross-selling strategies may improve customer retention and increase customer lifetime value.

CHURN BY ACTIVITY STATUS



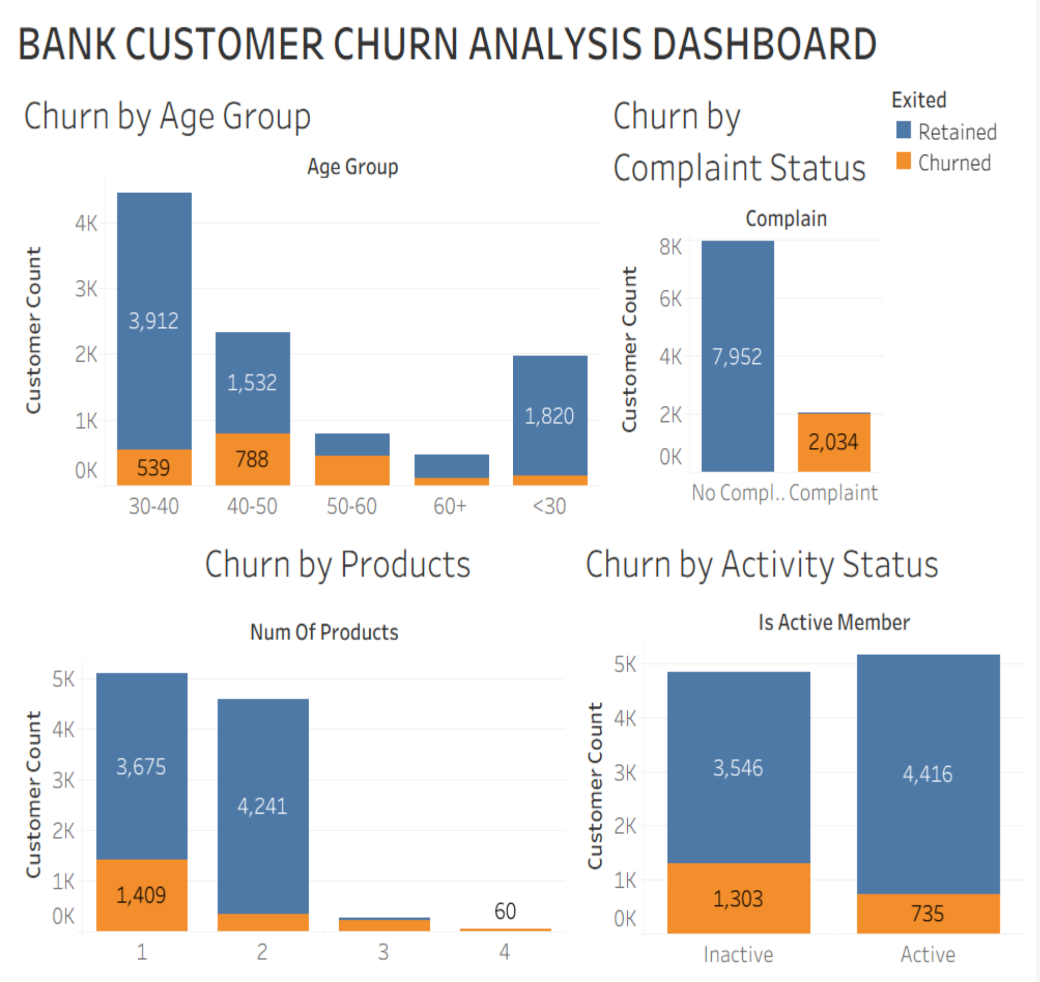
Key Findings

- **Inactive customers recorded 1,303 churn cases**, compared to **735 among active customers**.
- Active customers exhibited significantly higher retention (**4,416 retained customers**) than inactive customers (**3,546 retained customers**).
- Inactive customers are nearly **twice as likely to churn** as active customers.
- Customer engagement shows a strong relationship with retention outcomes.

Insight

Customer engagement is a critical driver of retention. Increasing customer interaction through personalized services, loyalty programs, and regular account activity may help reduce churn among inactive customers..

INTERACTIVE CUSTOMER CHURN DASHBOARD



Key Insights

- Customers with complaints exhibit the highest churn rates.
- Customers aged 50–60 are more likely to churn.
- Inactive customers show significantly higher churn levels.
- Customers with multiple products tend to have stronger retention.

Insight

The dashboard transforms complex customer data into actionable insights, enabling the bank to identify churn risks and improve retention strategies.

RECOMMENDATION

Recommendations

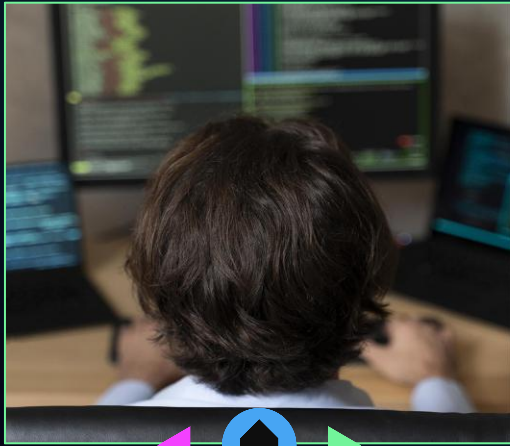
- Strengthen complaint resolution processes to reduce customer dissatisfaction.
- Develop targeted retention campaigns for customers aged 50 and above.
- Increase engagement programs for inactive customers.
- Promote cross-selling initiatives to encourage multi-product adoption.
- Deploy predictive models to identify customers at risk of churning.

Expected Impact

- Improved customer satisfaction.
- Higher customer retention rates.
- Increased product adoption.
- Reduced revenue loss from customer attrition.

CONCLUSION

- The overall churn rate is **20.38%**.
- Complaint status is the strongest predictor of customer churn.
- Older and inactive customers are more likely to leave the bank.
- Customers with multiple products generally demonstrate stronger loyalty.
- Data-driven retention strategies can significantly reduce customer churn.



THANK YOU

Questions & Discussion